



AI to predict protein structures

The discovery could speed up the search for new treatments, along with other applications. <http://dgit.in/aug21-51>



Malaria Vaccine

Biontech plans to bring a Malaria vaccine to the market by the end of 2022. <http://dgit.in/aug21-52>

This ice bends!



Source: New Scientist via YouTube

A simple trick allows us to make springy, flexible ice, and use them just like wires

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Ice is brittle, something we can observe first hand by watching it break easily and crumble when it is struck. Researchers in the labs of Zhejiang University in Hangzhou, China, have made a breakthrough, developing really thin strands of ice that are springy and snap back to their original shape even when bent to extreme angles. The development promises to change our understanding of ice forever. The researchers published their findings in a paper titled A flexible and springy form of ice, in *Science*.

What makes ice brittle is its structure and formation, but this discovery takes that theory that we have about ice and flips it on its head. Small crystals of frozen water, put together, give shape to how we see ice around us, in

everything from ice cubes to glaciers. Water is one of the most anomalous compounds known, so this strange and utterly new behaviour of ice is not entirely surprising. Ice has been known to morph into different structures depending upon the conditions that it is exposed to, so that is also one reason why it is possible for such a bendable structure of ice to exist.

In theory, ice tends to break easily because the stiff structure of the ice crystals that make it up, have little or no tolerance. In the solid-state, the molecules of water interact with each other in a certain way that leads to water molecules to establish a bond with the four neighbouring molecules of water, forming a lattice structure. This is because of a slight build of charges, the oxygen atoms form hydrogen bonds with neighbouring hydrogen atoms, at a distance from from the two hydrogen atoms in covalent bonds with the oxygen atom. The bonds between water molecules formed are stiff, and the close packing of adjacent molecules is what makes ice brittle. These are because of the impurities embedded within the ice, structural imperfections, and micro-cracks that are developed in every ice structure, all making ice prone to breaking.

This new bendy ice, though, does not exhibit such behaviour because the ice crystals were just pure ice without the imperfections, allowing the formation of perfect crystal lattices, that did not tend to break when bent. The ice was created inside a circular, 3D printed temperature controlled chamber.

The team then created their own mini laboratory inside the chamber, consisting of many tiny machines which were used in the experiments. 2,000 volts of electricity were applied through a tungsten needle within the container, which had been cooled to -50 degrees Celsius. Water molecules that were being attracted to the needle started freezing in thin strands of ice, along the direction of a spark, a fraction of the thickness of the human hair. Within the chamber were cameras set up to capture the strands of ice, too thin to be visible to the naked eye. The ice strands grew slowly at the rate of 0.2 millimeters every second, on strands that measured a few micrometers apart. These were not ordinary ice crystals. The crystals that formed at the tip of the needle had repeating patterns, which were perfect and had a honeycomb-like structure which led to an increase in their elasticity. These fibres had extremely smooth surfaces, and were uniform in their cross sections.

The secret to what allowed the ice microfibrils to be flexible and bendy, is that they were just pure ice, without the imperfections that cause the crystal formation to be interrupted. Just as electronic components are made up of pure industrial grade silicon, these ice crystals were made up of pure water. According to the researchers, very small strands of ice microfibrils form naturally in snowflakes, which are made up of many overlapping ice crystals, instead of a